

Alkylene oxide polymerizations: identification of side reactions and by-products

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Abstract In this study microwave-assisted anionic ring opening polymerization (ROP) of alkylene oxides is reported. Low molar mass polymers of propylene oxide (PO), butylene oxide (BO) and hexylene oxide (HO) are synthesized by anionic ring opening using various monohydric alcohols as initiators. The monohydric alcohols and lower molar masses allowed for monitoring of different types of possible unwanted by-products and their sources. Liquid chromatography at critical conditions (LCCC) and liquid adsorption chromatography (LAC) are used to determine the presence of oligomers other than targeted. Finally, MALDI-TOF MS of the products have clearly shown the trends and extents of different types of side reactions that are possible in anionic ring opening of alkylene oxides. Propylene oxide is most vulnerable to chain transfer reactions compared to higher alkylene oxide.

Keywords Polyoxyalkylenes · Anionic ring opening polymerization · Microwave-assisted polymerizations · HPLC of polymers

Introduction

Polyoxyalkylene are very important class of polymers for numerous applications in modern day material science [1–3]. Polyoxyethylenes (PEO) are the lowest member of this group and are very hydrophilic in nature. The higher

polyoxyalkylenes have pendant alkyl group and their hydrophobicity increases with the length of the alkyl group (methyl, ethyl and butyl) on the polymer chain. The solubility of polymers is strongly affected by the length of the polymer chains. Fairly high molar masses of Polyoxypropylene (PPO) are insoluble in water at room temperature. The polymers of higher alkylene oxides such as polyoxybutylene (PBO) and polyoxyhexylene (PHO) are soluble only in non-polar organic solvents and un-conventional solvents like silicon oils [4, 5]. The differences in the polarity of various polymers of alkylene oxides can be exploited for synthesis of polyoxyalkylene block copolymers with completely different physical properties [6].

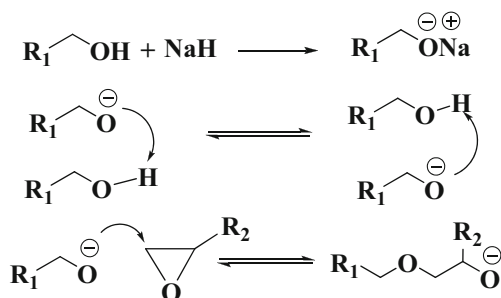
Different mechanisms reported for polymerization of alkylene oxides in literature are: coordinate-initiated, cationic or anionic [7]. Polymerization of PO has been discussed in detail in literature [8–13]. BO and HO have been polymerized using lithium as counter ion and tert butyl polyiminophosphazene base as complexing agent [4, 14], double metal cyanide (DMC) [15], and potassium alcoholate in combination with crown ethers as useful catalyst systems [5].

The cationic mechanism is not very favorable in the synthesis of block copolymers because of ring-chain equilibria (backbiting), which leads to the formation of products with a different architecture (linear and cyclic homopolymers to multiblocks). Coordinate-initiated polymerization is suitable for the production of homopolymers; in the synthesis of block copolymers, however, such catalysts may also yield chains with different functionalities.

Anionic ring opening polymerization (ROP) is the most widely employed technique for polymerization of alkylene oxides. Broadly accepted mechanism of anionic ROP of alkylene oxides is shown in Scheme 1. The problems associated with ROP of higher alkylene oxides by coordinate-initiated, cationic mechanisms can be avoided by anionic ROP,

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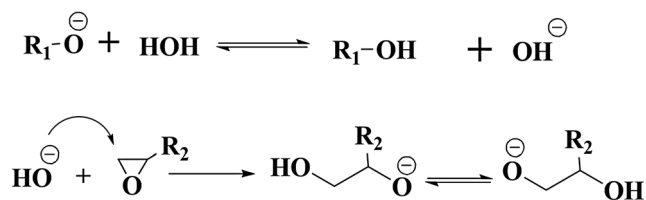
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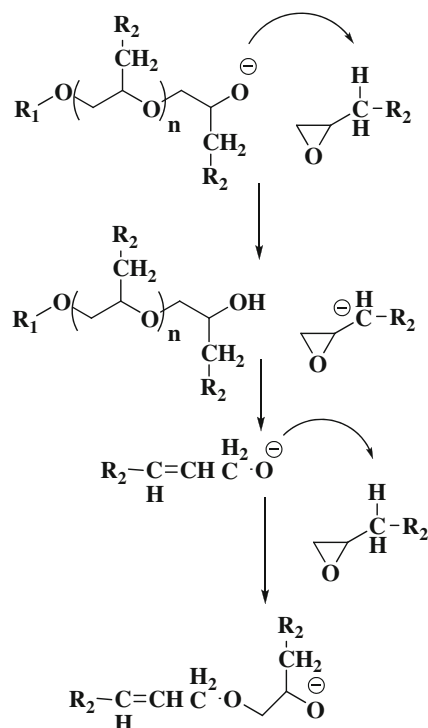
Scheme 1 Mechanism of anionic ring opening polymerization of alkylene oxides

however, there are other side reactions associated to it [5, 12, 13, 16, 17]. The lowest member of group (ethylene oxide-EO) can be polymerized by anionic ROP without any side reactions because of absence of any group for chain transfer reaction. However, higher alkylene oxides undergo several side reactions leading to homopolymer with unsaturation at the chain-end and homopolymer-diols. Homopolymer-diols originate from moisture in the reaction mixture as is shown in Scheme 2. An important side reaction may originate from chain transfer reaction to the monomer that resulted in the homopolymers with unsaturated end-group, Scheme 3. Another less probable side reaction could be chain transfer to the penultimate unit of the growing polymer chains that leads to the homopolymers with unsaturated end-group along with homopolymer diols, Scheme 4. Propylene oxide (PO) is most vulnerable to the chain transfer reactions because of high acidity of the methyl protons. The longer alkyl substituent on the oxirane ring may cause steric hindrance and these side reactions are less likely to happen for higher alkylene oxides. The presence of PPO homopolymers with allylic end-group along with PPGs and absence of PEO homopolymers in commercial poloxamers has been shown recently [18].

The nature of side reactions and their extent has been studied thoroughly for anionic ROP of PO [8–11, 13, 19]. However, current literature has not detailed information about anionic ROP of higher alkylene oxides (butylene oxide-BO and hexylene oxide-HO). There are only few reports about nature of side reactions in anionic ROP of higher alkylene oxides, especially lacking is any proof about functionality of side products, extent and probability of different type of side reactions which could be very different from PO because of longer substituent on the oxirane ring [5, 17].



Scheme 2 Homopolymer-diols originated from water content in reaction mixture

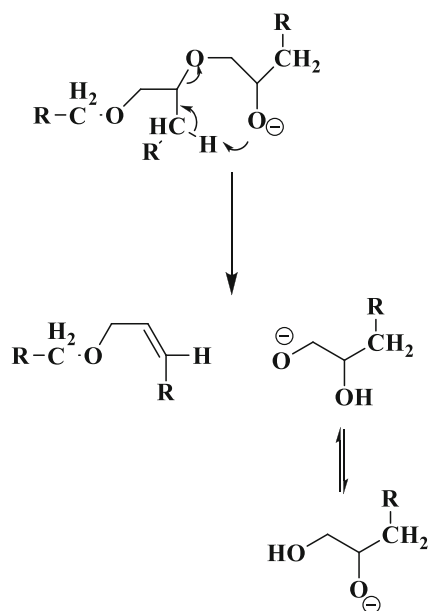


Scheme 3 Chain transfer to monomer leading to homopolymers with unsaturated end-group

The target of current systematic study is to determine the probability of different types of side reactions in anionic ROP of PO, BO and HO. The products are synthesized by microwave-assisted anionic ROP of alkylene oxides using different monohydric alcohols as initiators. Different modes of liquid chromatography of polymers (Liquid chromatography at critical conditions (LCCC) and liquid adsorption chromatography (LAC), and MALDI-TOF MS are used for detailed analysis of targeted and by-products. This study would help in better understanding of anionic ROP of alkylene oxides that can be used for synthesis of more complex architectures for several applications.

HPLC of polymers

The details of different modes of liquid chromatography of polymers can be found in recent reviews [20, 21]. Briefly, the interaction parameter ‘c’ is used to describe HPLC of polymers. If the value of the interaction parameter is negative for a certain polymer chain, the polymer should elute in SEC order, i.e. retention decreases with increasing molecular mass. A positive value of ‘c’ corresponds to the interaction/adsorption regime (retention increases exponentially with the number of repeat units). At the transition point of SEC and LAC, there is a point where the value of interaction parameter should be equal to zero. This point is known as critical adsorption point (CAP) and the corresponding polymer becomes “chromatographically invisible” at the specific mobile phase.



Scheme 4 Chain transfer to the penultimate unit of the growing polymer chains that forms homopolymer with unsaturated end-group along with homopolymer-diols

stationary phase-temperature system. Polymers at **CAP** elute as a sharp peak near the void volume of the column irrespective of their molar mass, but other distributions such as other block or functionality.

In the case of block polymers the interaction parameters c_A and c_B of the individual structural units (A and B) may have different values. At the CAP of the repeat unit A, the elution volume is independent of the chain length of the block A. In this case, AB polymers can be separated according to the block B.

Functional polymers can be separated by LCCC with regard to end-groups: all chains with the same end-group elute as a narrow peak at the same volume. The elution volume in this case depends upon the strength of the interaction of the end-group with the stationary phase.

In case of liquid adsorption chromatography (LAC), repeat unit has positive value of interaction parameter and elute after the void volume of the column in order of increasing number of repeat units. If end-group has stronger interaction with the stationary phase compared to the repeat unit, different homologous series could be separated from each other along with baseline separation of the individual oligomers of each homologous series.

Experimental

Polymerizations

Propylene oxide (PO), Butylene oxide (BO), hexylene oxide (HO) were purchased from Fluka (Buchs, Switzerland) and were refluxed over Calcium hydride prior to use.

Sodium hydride (60 % dispersion in mineral oil) was purchased from Aldrich. All alcohols like butanol, hexanol, octanol, decanol, dodecanol, tetradecanol, hexadecanol and octadecanol were purchased from Fluka (Buchs, Switzerland). Lower alcohols (butanol – hexanol) were dried over anhydrous sodium sulfate while higher alcohols (dodecanol, tetradecanol, hexadecanol and octadecanol) were used as received.

All polymerization were performed by microwave-assisted ring opening polymerizations of alkylene oxides as described in detail elsewhere [12, 13, 17].

Briefly the polymerization of alkylene oxides was performed by anionic ring opening using an alcohol and sodium hydride as initiator and co-initiator respectively. All polymerizations were performed in 2–5 mL (filling volume) microwave process vials (Biotage AB), which were oven dried under vacuum prior to use. Initiator and NaH were added to the vials which were subsequently sealed with an aluminum crimp top and Teflon septum. After purging the vials with argon for 5 min, the calculated volume of monomer was added with a syringe through the septum. The sealed microwave process vials were subsequently introduced into the cavity of the single-mode microwave reactor (Biotage Initiator Eight EXP 2.0, absorbance level: high). The temperature was kept 160 °C for PO, 180 °C for BO, and 200 °C for HO. Total reaction time varied from 1 to 4 h depending upon monomer and intended length of the chain. After polymerization, the active chain end was neutralized by adding an equimolar quantity of acetic acid.

Chromatography

These investigations were performed using the density detector (according to the mechanical oscillator principle) DDS70 (CHROMTECH, Graz, Austria). Data acquisition and processing was performed using the software CHROMA, which has been developed for the DDS70. In reversed phase LC, the mobile phase was delivered by a JASCO 880 PU pump (from Japan Spectroscopic Company, Tokyo, Japan) at a flow rate of $0.5 \text{ mL} \cdot \text{min}^{-1}$. Samples were injected manually using a Rheodyne 7125 injection valve (from Rheodyne, Cotati, CA, USA) equipped with an $50 \text{ } \mu\text{L}$ loop. A Waters 2414 refractive index detector (Waters, Milford, MA, USA) was connected to the DDS 70 for LAC separations. In case of LCCC, RI detector was replaced by a SEDEX 45 ELSD (Sedere, France). Nitrogen was used as carrier gas, and the pressure at the nebulizer was set to 1.0 bar. Evaporator temperature: $30 \text{ }^{\circ}\text{C}$.

In both systems, the columns and density cells were placed in a thermostatted box, in which a temperature of 25 °C was maintained.

Columns were connected to two column selection valves (Rheodyne 7060, from Rheodyne, Cotati, CA, USA).

Following reversed phase columns are used in this study

- **Prodigy ODS3** (Phenomenex, Torrance, CA, USA): silica-based octadecyl phase; 250×4.6 mm; particle diameter = $5 \mu\text{m}$; pore size = $100 \text{ \AA}$
- **Spherisorb S5X** (PhaseSeparations) C18; 250×4.6 mm; particle diameter = $5 \mu\text{m}$; pore size = $300 \text{ \AA}$
- **Jordi Gel DVB 500 RP**: 100 % poly(divinylbenzene); 250×4.6 mm; particle diameter = $5 \mu\text{m}$; nominal pore size = $500 \text{ \AA}$, (Jordi, Bellingham, MA, USA)

Mobile phases were mixed by mass and vacuum degassed, their composition was controlled by density measurement using a DMA 60 density meter equipped with a measuring cell DMA 602 M (A. Paar, Graz, Austria).

The solvents (THF, acetone and water, all HPLC grade) were purchased from Roth (Karlsruhe, Germany) and Merck (Darmstadt, Germany).

Spectroscopy

MALDI-TOF Mass Spectrometry was performed on a Micromass TofSpec 2E Time-of-Flight Mass Spectrometer. The instrument is equipped with a nitrogen laser (337 nm wavelength, operated at a frequency of 5 Hz), and a time lag focusing unit. Ions were generated by irradiation slightly above the threshold laser power. Positive ion spectra were recorded in reflectron mode applying an accelerating voltage of 20 kV and externally calibrated with a suitable mixture of poly(ethyleneglycol)s (PEG). The spectra of 100–150 shots were averaged to improve the signal-to-noise ratio. Analysis of data was done with MassLynx-Software V3.5 (Micromass/Waters, Manchester, UK).

Sample were prepared by mixing a solution of dithranol ($c = 10 \text{ mg/mL}$ in THF), a solution of the analyte ($c = 2 \text{ mg/mL}$ in THF), and a solution of CF_3COONa (NaTFA; $c = 0.1 \text{ mg/mL}$ in THF), in a ratio of 7:1:1 (v/v/v). $0.5 \mu\text{L}$ of the resulting mixture were deposited on the sample plate (stainless steel) and allowed to dry under air.

Results and discussion

Anionic ring opening polymerization of alkylene oxides is associated with several different types of side reactions that lead to by-products other than targeted, see Schemes 1, 2, 3, and 4. These side reactions are not possible in ring opening of EO due to absence of any group for chain transfer reaction, therefore, fairly clean PEOs are available in the market. PO has a methyl group on the oxirane ring that is vulnerable to chain transfer because of its high acidity. In our recent paper about comprehensive analysis of commercial poloxamers, it has been shown that significant amounts of PPOs with allylic

end-group and PPGs (lesser extent) are present whereas no indication of PEO homopolymers is found in the samples [18]. The reaction mechanism of polymerization of higher alkylene oxides should be same but extent of side reactions could be different due to presence of pendant groups of different lengths. The chain transfer reaction leading to homopolymers with unsaturation at one end is less likely to occur in polymerization of higher alkylene oxides (BO and HO) due to steric hindrance of longer alkyl group, Scheme 3 [17, 22, 23]. The other side reaction initiated with the moisture content in the reaction mixture can be avoided or at least minimized by carefully drying the reagents, Scheme 2. The probability of the side reaction shown in Scheme 4 is less compared to others (Schemes 2 and 3). To study effect of the length of the pendant group on oxirane ring on the extent of side reactions, we polymerized PO (methyl pendant group), BO (ethyl pendant group) and HO (butyl pendant group) at their optimum reaction conditions [12, 13, 17]. The target of this study is to identify all the side products formed during anionic ROP of alkylene oxides. For this purpose, different alcohols are used as initiators to differentiate targeted product from unavoidable by-products. In the following discussion, we present various complimentary approaches to analyze targeted and unavoidable by-products of alkylene oxide polymerization.

Liquid adsorption chromatography (LAC)

If a fatty alcohol is used as initiator, the targeted product should be polyalkylene oxide with corresponding alkyl end-group. In RPLC, longer alkyl end-group should adsorb strongly compared to the repeat unit and elution volume of oligomers would be higher. The elution volume of oligomers should be representative of adsorption strength of repeat unit and end-groups. Therefore, the homologous series having an alkyl end-group should elute later than the corresponding alcohol. The other by-products that do not have strongly adsorbing end-group should elute earlier than the alcohols, at least for lower molar masses. This approach is used to determine the extent of by-products in the samples. Figure 1 demonstrates the elution profiles of polyoxyalkylenes with end-groups of different adsorption strength and corresponding alcohols. PPG 425 and PPO with allylic end-groups eluted earlier than decanol, furthermore, oligomers of decanol-based PPOs eluted later than decanol. The decanol-based product also contains some PPOs that eluted earlier than decanol and are supposed to be by-products as described earlier, Fig. 1a. Figure 1b, c depict the elution profiles of PBOs and PHOs. As can be seen the traces of oligomers eluting earlier than corresponding alcohols are found. The amount of the by-products seems to be significantly low compared to PPOs. The targeted products eluted after the corresponding alcohol whereas by-products eluted earlier as a different series. Since RI is used as detector for this analysis, a reasonable idea of relative amounts of different homologous

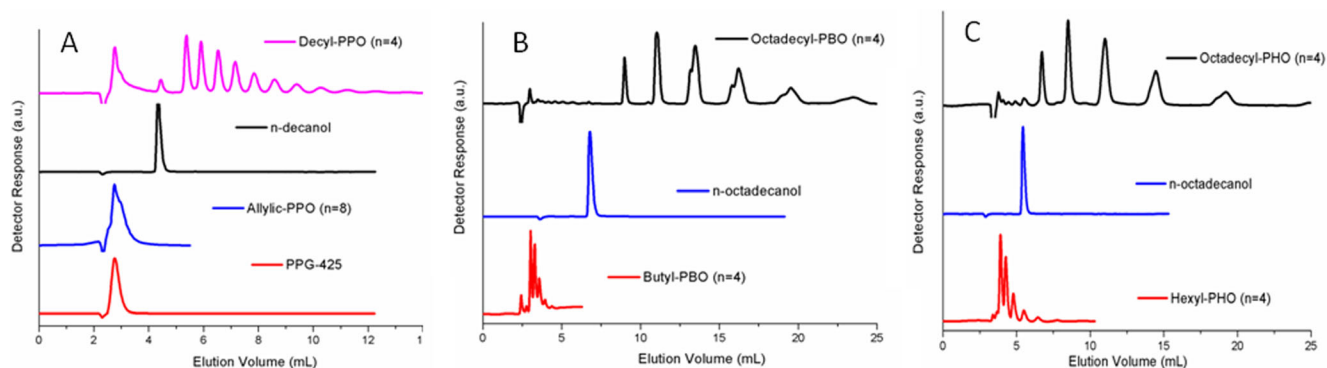


Fig. 1 LAC separation of polyoxyalkylenes **a)** PPOs on Prodigy ODS3 250×4.6 , $100\text{\AA}$ in acetone/water = 80/20 w/w; **b)** PBOs on Prodigy ODS3 250×4.6 , $100\text{\AA}$ in acetone/water = 90/10 w/w; **c)** PHOs on Spherisorb S5X C18, $300\text{\AA}$ in acetone/water = 90/10 w/w

series is obtained. Differentiation between polyoxyalkylene-diols from chains with unsaturated end-group is not possible by this approach.

Liquid chromatography at critical conditions (LCCC)

Liquid chromatography at critical conditions (LCCC) is a powerful technique to separate polymers according to their functionality. Critical conditions for PPO were found in 80 % THF: Water mobile phase on Jordi RP column [12, 24, 25]. Figure 2a depicts the separation of PPO chains with different end-groups. PPGs eluted at 2.62 mL while PPO chains with allylic end-group eluted at 2.85 mL. The PPO chains with octadecyl end-group eluted at 3.34 mL. The octadecyl-PPO contains significant amounts of PPGs and PPOs with allylic end-group that are separated excellently from the targeted product. It is evident from figure that major side reaction occurring in anionic ROP of PO is chain transfer to monomer, Scheme 3. The octadecyl-based PPO was fractionated into two parts for MALDI-TOF MS analysis. Fraction limits are from 2.0 to 3.0 mL (F-1) and 3.0–4.0 mL (F-2), of chromatogram of octadecyl-PPO as shown in Fig. 2a.

Critical adsorption point (CAP) of PBO and PHO are established on the same column in 84 and 87 % THF, respectively [17, 22, 23, 25]. In case of PBO, the unsaturated end-

group has similar selectivity as that of repeat unit and it is not possible to differentiate PBO-diols from PBOs with unsaturated end-group. At CAP of PBO, butyl-PBO eluted at 2.98 mL. The peak has a shoulder in front that might be due to PBO-diols. To detect the by-products, octadecanol-based product is also analyzed at CAP of PBO. As can be seen, octadecyl-PBOs eluted at 3.05 mL. The sample has small amounts of by-products that has different functionality than the targeted product and eluted earlier (2.98 mL). The early eluting peak is attributed to the by-products formed during ROP of BO, Schemes 1, 2, 3 and 4. Due to unavailability of reasonable standards, it is difficult to conclude about functionality of the by-products. The CAP for PHO was found in 87 % THF-water on the same column. HO repeat unit is very non-polar. Hence, addition of another non-polar group does not have much effect on the chromatographic retention in LCCC, Fig. 2c. Octadecanol-based PHO eluted slightly later than hexanol-based PHO. It also has a slight shoulder at lower elution volume, which corresponds to by-products formed during polymerization. LCCC allowed for separation of by-products from targeted products with only difference of end-groups. It works excellently for PPOs but its efficiency decreases as the length of pendant group on the polymer chain increased (from methyl to butyl), probably due to decreasing difference of polarity between repeat unit and end-group.

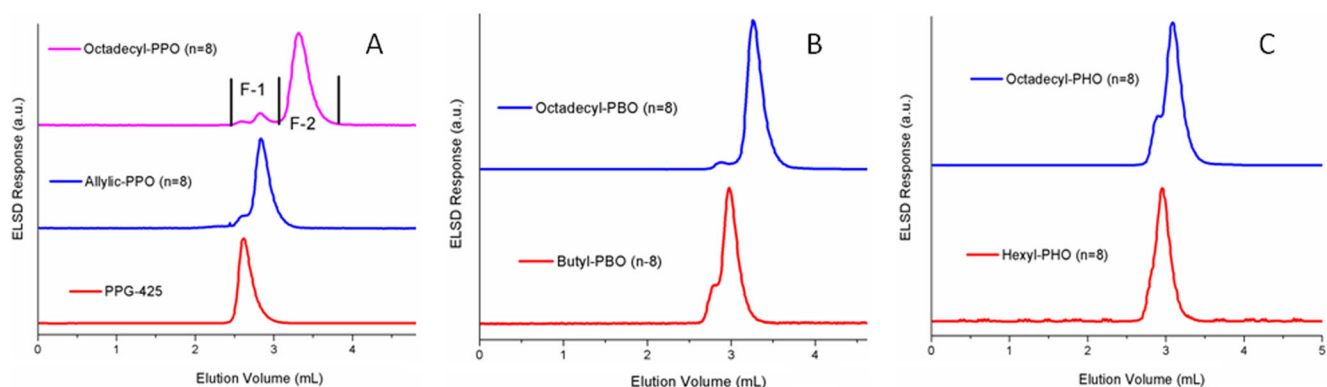
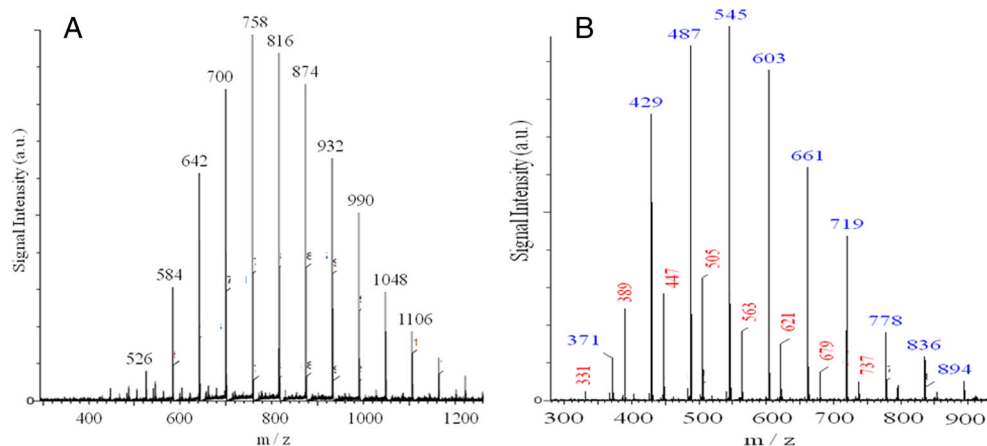


Fig. 2 Liquid chromatography at critical conditions on Jordi DVB 500 RP columns in THF/Water mobile phase system, **a)** PPO in THF/Water = 80/20 w/w, F-1 and F-2 show the fractionation limits; **b)** PBO in THF/Water = 84/16 w/w; **c)** PHO in THF/Water = 87/13 w/w

Fig. 3 MALDI-TOF mass spectra, **a)** octadecyl-PPO, F-2 (3.0–4.0 mL, Fig. 2a); **b)** octadecyl-PPO, F-1 (2.0–3.0 mL, Fig. 2a); Blue label (major series): allylic-PPOs, Red label (minor series): PPGs; Theoretical molecular weight of Octadecyl-P(PO)₉ (C₄₅H₉₂O₁₀Na) = 816.21, monoallyl-P(PO)₈ (C₂₇H₅₄O₉Na) = 545.36 and PPGs (PO)₉ (C₂₄H₅₀O₉Na) = 505.33



MALDI-TOF MS

MALDI-TOF MS is obviously the best technique to have a better picture of the by-products having similar chemical nature but different end-groups, see Schemes 1, 2, 3 and 4. The results by chromatography are really endorsed by spectroscopy along with other useful additional information.

As mentioned earlier, octadecyl-PPO was fractionated at critical conditions of PPO. The fractions are subjected to MALDI-TO MS analysis. Figure 3a depicts the MALDI-TOF mass spectra of fraction 2 (F-2). Evidently, it is a clean fraction of targeted octadecyl-PPOs. The other fraction (F-1) consists of two series of peaks, Fig. 3b. Major series belong to PPOs with unsaturation at the chain end that is possible by chain transfer to the monomer, Scheme 3. Minor series are PPGs that are formed by initiation of polymerization with the moisture in the reaction mixture, Scheme 2. The MALDI-TOF MS results endorsed chromatographic results that there are two types of end-groups of the by-products that are initiated by different mechanism as shown in Schemes 1, 2, 3 and 4.

MALDI-TOF MS analysis of raw PBOs and PHOs initiated by different alcohols is shown in Figs. 4 and 5, respectively. Butyl-PBO has two series of peaks. Major series belong to the targeted product. In this case there is only one minor series that is PBO-diols, Fig. 4a. Octadecyl-PBO also have two series of oligomers, the major series is the targeted product initiated with octadecanol whereas minor series is PBO-diols that are formed by initiation with the moisture, Fig. 4b.

Similar trend continues for both hexanol and octadecanol-based PHOs. Only by-product in the samples was PHO-diol that is formed by initiation with moisture. Therefore, hypothesis at the start of the discussion that chain transfer to monomer is less likely to occur with longer pendant alkyl group (ethyl and butyl) on the oxirane ring, seems to be correct. PO is most vulnerable to chain transfer to the monomer due to high acidity of the methyl group. Therefore, it is pertinent to say that chain transfer to the monomer or penultimate chain is most probable side reaction in anionic ROP of PO while the increasing length of substituent (ethyl and butyl) on the oxirane ring hinders the chain transfer reaction in ROP of higher alkylene oxides (BO and HO). The results obtained by three different techniques

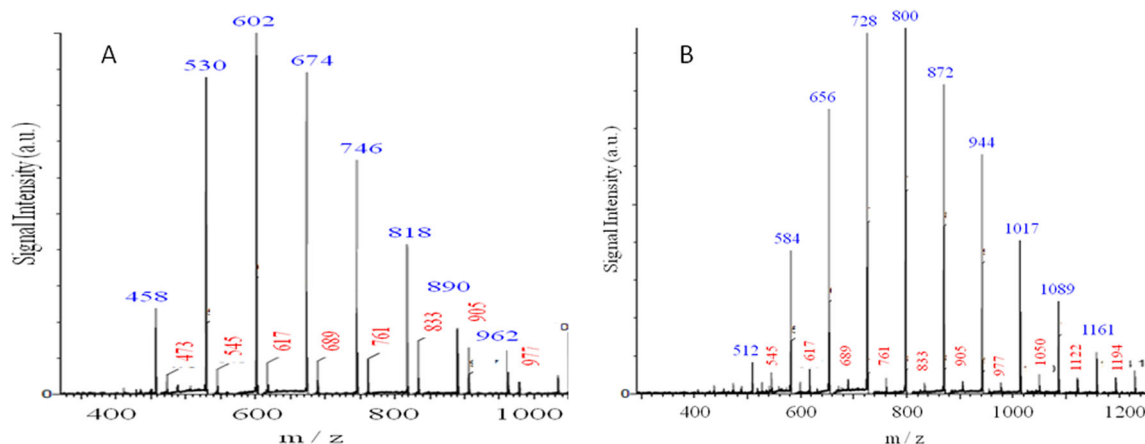


Fig. 4 MALDI-TOF mass spectra, **a)** butyl-PBO; **b)** octadecyl-PBO; Blue label (major series): targeted product, Red label (minor series): PBO-diols; Theoretical molecular weight of butyl-P(BO)₇ (C₃₂H₆₆O₈

Na) = 601.86, P(BO)₈-diol (C₃₂H₆₅O₉Na) = 616.86 and octadecyl-P(BO)₇ (C₄₆H₉₆O₈Na) = 800.25

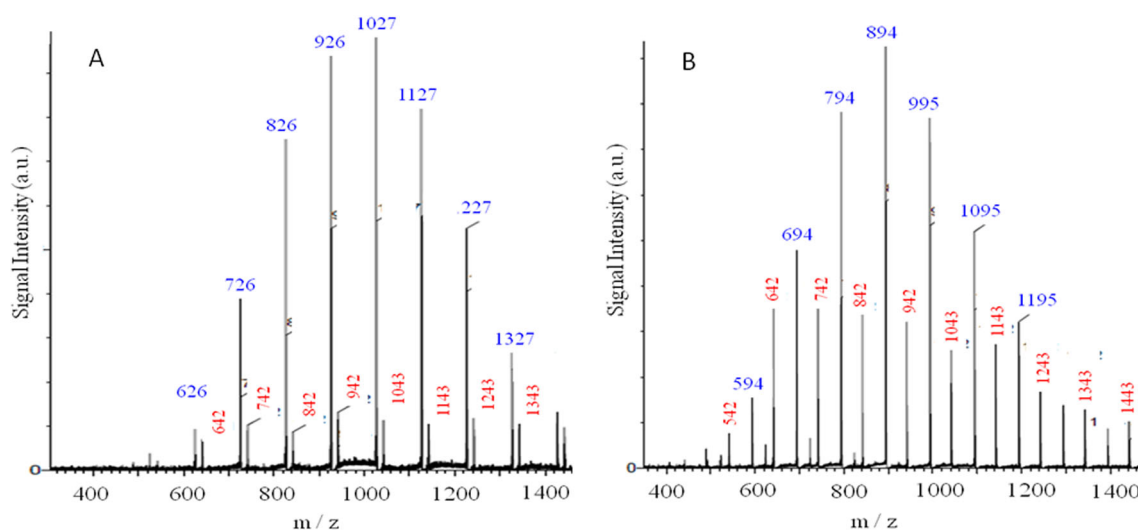


Fig. 5 MALDI-TOF mass spectra, **a)** hexyl-PHO; **b)** octadecyl-PHO; Blue label (major series): targeted product, Red label (minor series): PHO-diols; Theoretical molecular weight of hexyl-P(HO)₁₀ (C₆₆ H₁₃₄

O₁₁ Na) = 1126.78, P(HO)₉-diol (C₅₄ H₁₁₀ O₁₀ Na) = 942.45 and octadecyl-P (HO)₆ (C₅₄ H₁₁₀ O₇ Na) = 894.45

synergistically provide important information on anionic ring opening polymerization of alkylene oxides.

Conclusion

Extent of various side reactions in anionic ROP of alkylene oxides depends strongly upon the length of pendant group on the oxirane ring. ROP of PO is accompanied by chain transfer reaction leading to polymer chains with unsaturated end-group. On the other hand, chain transfer to monomer is less likely to happen for higher alkylene oxides (BO and HO) due to steric hindrance caused by longer pendant groups (ethyl and butyl) on the oxirane ring. In the latter case, only homopolymer-diols were detected that are result of initiation of polymerization by moisture content in the reaction mixture. Combined results of chromatography and spectroscopy provided more detailed information on targeted products and possible routes of formation of unwanted by-products.

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